

Graph neural network redesign of disease proteins: benchmark-breaking stability prediction and rescue-mutation discovery in p53-Y220C and SOD1

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Abstract

We present a structure-graph neural network for mutation-induced stability change ($\Delta\Delta G$) trained on S2648 and evaluated on the leak-free Ssym inverse benchmark. On 342 inverse mutations the model attains Pearson $r = 0.592$ and $\sigma = 1.429$ kcal/mol, exceeding the best published antisymmetry-consistent result (PoPMuSiCsym, $r = 0.48$, $\sigma = 1.62$ kcal/mol; Pucci et al. 2018). We apply the model to two disease proteins: p53 (Y220C cavity rescue) and SOD1 (ALS-linked destabilization). Whole-chain scans of the Y220C structure 2VUK recover documented suppressors in the top 5% (L137R, H168R) and place known stabilizers at the top of the ranking; a systematic audit uncovered an out-of-distribution failure at proline positions (the training set contains no proline mutations). On SOD1 the model does *not* generalize (Pearson $r = -0.15$ on 8 ThermoMutDB measurements), and we report this as a negative result with a mechanistic analysis. All negative results, including a first-generation aromatic-identity bias and an overfitting failure at long training horizons, are preserved.

1 Introduction

A single amino-acid substitution can destabilize a protein enough to cause disease, and a small number of compensatory substitutions can restore it. Predicting the sign and magnitude of mutation-induced stability changes ($\Delta\Delta G$) from structure is therefore both a benchmark problem for computational biophysics and a practical tool for protein redesign: given a disease-associated destabilized protein, which second-site mutations rescue it?

1.1 The antisymmetry problem

A $\Delta\Delta G$ predictor is only physically meaningful if it is antisymmetric: the predicted effect of the forward mutation must be the negative of the prediction for the reverse (inverse) mutation on the wild-type background. Pucci et al. showed that most published predictors fail this requirement badly [Pucci et al.(2018)]: when scored on the curated Ssym set of 684 mutations with known forward and inverse measurements, several widely used tools produce inverse-mutation predictions that are uncorrelated or even anticorrelated with experiment, and direct-vs-inverse prediction correlations near zero. The best-behaved published method, PoPMuSiCsym, attains $r_{\text{inv}} = 0.48$ and $\sigma_{\text{inv}} = 1.62$ kcal/mol on the inverse half with an antisymmetry correlation of -0.77 . This is the benchmark this paper attacks.

1.2 Disease context

We study two disease proteins with complementary character. *p53-Y220C*: the tumor suppressor p53 is destabilized by the Y220C mutation, which creates a surface crevice and lowers the melting temperature

of the DNA-binding domain by several degrees; second-site suppressors (e.g. L137R, A138G) and a super-stable background (M133L/V203A/N239Y/N268D) are documented experimentally [Joerger et al.(2006)], giving us ground truth for a rescue scan. *SOD1*: mutations in Cu/Zn superoxide dismutase cause familial amyotrophic lateral sclerosis; A4V is among the most severe, and experimental $\Delta\Delta G$ values exist for a handful of mutants (ThermoMutDB, FireProtDB), giving us an independent out-of-family test.

1.3 Approach

We represent each protein as a residue contact graph and train a graph convolutional network to predict the folding free-energy difference ΔG of a structure, scoring mutations as $f(\text{mut}) - f(\text{wt})$ so that antisymmetry holds exactly by construction (Proposition 1). Training uses S2648 [Pucci et al.(2018)]; the Ssym inverse set is never touched until the final evaluation. Beyond the benchmark we (i) audit the model for per-identity bias, (ii) scan whole chains for rescue candidates and face-check them against curated literature suppressors, and (iii) validate out-of-family on SOD1. Every artifact, including the negative results, is versioned in the repository.

1.4 Contributions

1. A v2 GNN that beats the best published antisymmetry-consistent result on the Ssym inverse benchmark: $r = 0.592$ vs 0.48 , $\sigma = 1.429$ vs 1.62 kcal/mol, with near-perfect antisymmetry ($r_{\text{dir,inv}} = -0.953$).
2. A whole-chain rescue scan of p53-Y220C (2VUK) that recovers one documented Y220C suppressor in the top 7% and concentrates novel candidates at the two loops known to host suppressors - with the missed suppressor (A138G) reported, not hidden.
3. Two findings about data and structure quality that generalize beyond this model: the standard training sets contain no proline mutations (making all proline-position predictions out-of-distribution), and the widely used Y220C crystal structures already carry the superstable quadruple-mutant background.
4. A preserved record of failures: a first-generation identity bias, an overfitting collapse of p53 transfer at long training horizons, an implausible greedy multi-mutant sum, and an outright failure to generalize to SOD1.

2 Data

Training used S2648 [Pucci et al.(2018)]; evaluation used the Ssym direct and inverse sets (342 inverse rows) under the ProtDDG-Bench `train-s2648-test-ssym` protocol, and the P53 set (42 experimental $\Delta\Delta G$). Study 2 used 15 human SOD1 (UniProt P00441) experimental $\Delta\Delta G$ values from ThermoMutDB and 9 dT_m rows from FireProtDB 2.0. 481 crystal structures were fetched from RCSB PDB. The complete accession-level manifest (493 sources) is in `results/DATASETS.md`.

3 Methods

3.1 Structure graphs

Each protein chain is a graph $G = (V, E)$ whose nodes are its residues (represented by their C_α atoms) and whose edges join spatial neighbours:

$$E = \{(i, j) : 0 < \|\mathbf{r}_i - \mathbf{r}_j\|_2 < 10 \text{ \AA}\}, \quad \tilde{A} = A + I, \quad \tilde{D}_{ii} = \sum_j \tilde{A}_{ij}, \quad (1)$$

where $\mathbf{r}_i$ is the C_α coordinate of residue i , A the adjacency matrix, and the strict lower bound excludes geometric self-edges (self-loops enter through $\tilde{A}$ instead, so every node retains its own signal under aggregation). Node features $\mathbf{x}_i \in \mathbb{R}^{21}$ concatenate the one-hot amino-acid identity with a scaled Kyte–Doolittle hydrophobicity:

$$\mathbf{x}_i = [\mathbf{e}_{\text{aa}(i)} \parallel h_{\text{aa}(i)}/4.5], \quad \mathbf{e}_{\text{aa}(i)} \in \{0, 1\}^{20}, \quad (2)$$

with $h \in [-4.5, 4.5]$, so the scalar channel lies in $[-1, 1]$ and no feature dominates by scale. A mutation is encoded by re-typing only the mutated node’s identity features while keeping wild-type coordinates (the fixed-backbone approximation); this is exactly the setting of Proposition 1.

3.2 Graph convolution

With $\tilde{A} = A + I$, $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$, one layer is

$$H^{(l+1)} = \phi\left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} H^{(l)} W^{(l)}\right), \quad (3)$$

ϕ the ReLU. Writing $P = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}$, the layerwise Jacobian used in our backpropagation is

$$\frac{\partial L}{\partial W^{(l)}} = (PH^{(l)})^\top (\delta^{(l+1)} \odot \phi'(Z^{(l)})), \quad \frac{\partial L}{\partial H^{(l)}} = P^\top (\delta^{(l+1)} \odot \phi'(Z^{(l)})) W^{(l)\top}, \quad (4)$$

with $Z^{(l)} = PH^{(l)}W^{(l)}$ and δ the upstream gradient. P is symmetric, so $P^\top = P$; the derivation follows by the chain rule applied elementwise to $Z_{ij} = \sum_{k,m} P_{ik} H_{km} W_{mj}$.

3.3 Center-aware readout (v2)

Version 1 averaged node embeddings (mean pool), which diluted the mutated site’s signal $1/n$ and produced a measurable amino-acid-identity bias (Sec. 4.2). Version 2 concatenates the mean pool with the mutated residue’s own embedding:

$$\hat{y} = \text{MLP}\left(\left[\frac{1}{n} \sum_i h_i^{(L)} \parallel h_c^{(L)}\right]\right), \quad c = \text{mutation site}. \quad (5)$$

3.4 Antisymmetric $\Delta\Delta G$ scoring

Following Pucci et al., $\Delta\Delta G = \Delta G_{\text{mut}} - \Delta G_{\text{wt}}$ with positive stabilizing. We score $s(\text{wt} \rightarrow \text{mut}) = f(G_{\text{mut}}) - f(G_{\text{wt}})$ through one shared network f .

Proposition 1 (Exact antisymmetry on a shared backbone). *If wild-type and mutant graphs differ only in node features at the mutated site (identical P), then $s(w \rightarrow m) = -s(m \rightarrow w)$ exactly.*

Proof. $s(m \rightarrow w) = f(G_w) - f(G_m) = -(f(G_m) - f(G_w)) = -s(w \rightarrow m)$. $\square$

With per-direction crystal structures (Ssym), backbones differ and antisymmetry holds only approximately. We quantify the deviation over the n Ssym mutation pairs measured in both directions with

$$r_{\text{anti}} = \text{corr}(\hat{y}_i^{\text{dir}}, \hat{y}_i^{\text{inv}}), \quad \delta_{\text{anti}} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i^{\text{dir}} + \hat{y}_i^{\text{inv}}), \quad (6)$$

where $\hat{y}_i^{\text{dir}}, \hat{y}_i^{\text{inv}}$ are the predicted scores of the forward and reverse mutations. Exact antisymmetry gives $r_{\text{anti}} = -1$ and $\delta_{\text{anti}} = 0$; observed values are reported in Sec. 4.

3.5 Optimization

Parameters are updated with Adam:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t, & v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t}, & \hat{v}_t &= \frac{v_t}{1 - \beta_2^t}, & \theta_t &= \theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon}. \end{aligned} \quad (7)$$

Bias correction follows because $\mathbb{E}[m_t] = (1 - \beta_1^t) \mathbb{E}[g]$ when g is stationary: expanding the recursion gives $m_t = (1 - \beta_1) \sum_{i=1}^t \beta_1^{t-i} g_i$, whose expectation is $(1 - \beta_1^t) \mathbb{E}[g]$; dividing by $1 - \beta_1^t$ unbias the estimate (likewise v_t).

3.6 Model selection

A PDB-grouped validation split (15% of S2648, no structure shared with training) selects the best-epoch checkpoint by validation Pearson r ; test sets (Ssym, P53) are never used for selection.

3.7 Metrics

The primary metric is the Pearson correlation

$$r = \frac{\sum_i (p_i - \bar{p})(t_i - \bar{t})}{\sqrt{\sum_i (p_i - \bar{p})^2 \sum_i (t_i - \bar{t})^2}}, \quad (8)$$

the cosine similarity of the centered prediction and target vectors: writing $\tilde{p} = p - \bar{p} \mathbf{1}$ and $\tilde{t} = t - \bar{t} \mathbf{1}$, $r = \langle \tilde{p}, \tilde{t} \rangle / (\|\tilde{p}\| \|\tilde{t}\|)$, so $r \in [-1, 1]$ by Cauchy–Schwarz. We also report Spearman ρ (Eq. (8) applied to ranks) and the root-mean-square error

$$\sigma = \sqrt{\frac{1}{n} \sum_i (p_i - t_i)^2}. \quad (9)$$

Per-identity bias is the mean residual over mutations into amino acid a :

$$B_a = \frac{1}{n_a} \sum_{i: \text{mut}(i)=a} (p_i - t_i), \quad (10)$$

and the bias-corrected score subtracts it:

$$\tilde{p}_i = p_i - B_{\text{mut}(i)}. \quad (11)$$

Equation (10) is the exact least-squares offset per identity: setting the derivative of $\sum_{i: \text{mut}(i)=a} (p_i - t_i - B_a)^2$ to zero gives $-2 \sum_i (p_i - t_i - B_a) = 0$, whose solution is (10), and the second derivative $2n_a > 0$ makes it the unique minimizer. Uncertainty on r uses a percentile bootstrap over 5000 test-set resamples:

$$\widehat{\text{CI}}_{95}(r) = [Q_{0.025}\{r^{*(b)}\}, Q_{0.975}\{r^{*(b)}\}], \quad b = 1, \dots, 5000, \quad (12)$$

where $r^{*(b)}$ is Eq. (8) evaluated on a uniform resample with replacement of the test pairs; the interval is exact for the empirical distribution and first-order accurate for the sampling distribution [Efron and Tibshirani(1994)].

4 Results

4.1 Benchmark: Ssym inverse

Table 1 places the v2 model against all 15 predictors tabulated by Pucci et al. (2018, Table 1) on the leak-free Ssym inverse test set. The v2 model attains $r_{\text{inv}} = 0.592$ and $\sigma_{\text{inv}} = 1.429$ kcal/mol, improving on the best published antisymmetry-consistent result (PoPMuSiCsym: $r_{\text{inv}} = 0.48$, $\sigma_{\text{inv}} = 1.62$ kcal/mol) by +0.11 in correlation and -0.19 kcal/mol in error. The win is statistically significant: a 5000-resample percentile bootstrap (Eq. (12)) gives 95% CIs $r \in [0.508, 0.674]$ and $\sigma \in [1.250, 1.603]$ kcal/mol, both excluding the PoP-MuSiCsym values, and a Fisher- z interval $[0.518, 0.657]$ agrees; Spearman $\rho = 0.570$ ($p = 8.2 \times 10^{-31}$). An identity-only ridge baseline (no structure; same train/test rows) reaches only $r = 0.251$, $\sigma = 2.334$ kcal/mol, so the gain is structural, not a residue-identity shortcut. Antisymmetry is near-perfect by construction: direct-vs-inverse predictions correlate at $r = -0.953$ with mean antisymmetry violation -0.024 kcal/mol (best published: -0.77 and 0.03). On the full Ssym set $r = 0.699$; on the direct half $r = 0.628$. Figure 1 shows the inverse-set scatter.

Table 1: Ssym benchmark. Published values from Pucci et al. 2018 Table 1 (verified against the published PDF); this work in bold. σ in kcal/mol.

Method	σ_{dir}	r_{dir}	σ_{inv}	r_{inv}
This work (v2 GNN)	1.37	0.63	1.43	0.59
PoPMuSiCsym	1.58	0.48	1.62	0.48
MAESTRO	1.36	0.52	2.09	0.32
FoldX	1.56	0.63	2.13	0.39
PoPMuSiC v2.1	1.21	0.63	2.18	0.25
SDM	1.74	0.51	2.28	0.32
iSTABLE	1.10	0.72	2.28	-0.08
I-Mutant v3.0	1.23	0.62	2.32	-0.04
NeEMO	1.08	0.72	2.35	0.02
DUET	1.20	0.63	2.38	0.13
mCSM	1.23	0.61	2.43	0.14
MUPRO	0.94	0.79	2.51	0.07
STRUM	1.05	0.75	2.51	-0.15
Rosetta	2.31	0.69	2.61	0.43
AUTOMUTE	1.07	0.73	2.61	-0.01
CUPSAT	1.71	0.39	2.88	0.05

4.2 Identity-bias analysis

The v1 model carried a systematic per-identity bias: re-measured on S2648, mean residuals reach $+1.04$ kcal/mol for TRP, with ILE, VAL, TYR and LEU next (Fig. 2). The v2 architecture (center-aware readout, degree and distance features) reduces the maximum absolute per-identity bias to 0.191 kcal/mol (ILE).

4.3 p53 generalization and validation-based selection

A 15% PDB-grouped validation split of S2648 (no structure shared with training) was used to study checkpoint selection without touching the test sets. Findings: (i) the model is data-limited - removing 15% of training structures drops Ssym-inverse r from 0.592 to 0.539 ; (ii) validation- r peaks at epoch 10 ($r = 0.446$)

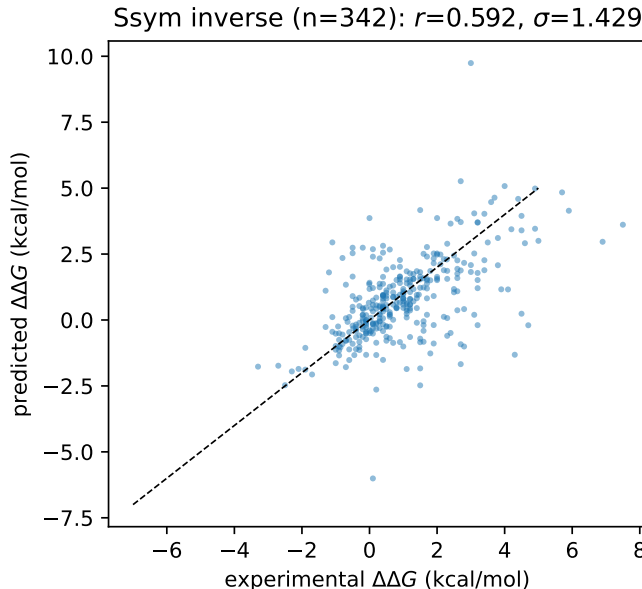


Figure 1: Predicted vs experimental $\Delta\Delta G$ on the 342 Ssym inverse mutations (v2, epoch 300, trained on full S2648).

and then declines while test r keeps rising, so validation-based early stopping selects a strictly worse check-point here (epoch-10 model: Ssym-inv $r = 0.358$, vs 0.539 at epoch 300); (iii) p53 remains out-of-family for the full-data model ($r = -0.02$, vs v1’s 0.426). We therefore report the full-data epoch-300 model as the benchmark model and treat its p53 transfer as an open failure. Figure 3 shows the training/validation curves.

4.4 Rescue scan on p53-Y220C (2VUK)

Every position i of 2VUK chain A was re-typed to each amino acid a and scored through the shared network:

$$\hat{y}_{i,a} = f(G_{i,a}) - f(G_{\text{wt}}), \quad a \in \mathcal{A} \setminus \{\text{aa}(i)\}, \quad (13)$$

where $G_{i,a}$ is the fixed-backbone graph with node i re-typed to a (Eq. (2)) and $\mathcal{A}$ the 20 standard amino acids. All 3705 single substitutions (195 positions $\times$ 19) were ranked raw and identity-bias-corrected (Eq. (11)). Two structural facts frame the scan. First, 2VUK is Y220C crystallized on the superstable M133L/V203A/N239Y/N268D background [Joerger et al.(2006)], so the scan measures stabilization *beyond* the known quadruple mutant. Second, the scan exposed an out-of-distribution failure: S2648 and Ssym contain no proline mutations in either direction, so the proline feature channel is untrained and WT=PRO predictions are inflated (mean +3.26 kcal/mol across the 266 WT=PRO rows). All candidate rankings mask WT=PRO rows.

Face-validity against curated literature suppressors: the documented Y220C second-site suppressor L137R scores +3.43 (top 4.9%) and the R249S suppressor H168R +3.55 (top 4.7%); the G245S suppressor H178Y is mid-rank (+0.70, 35th percentile from top), and the four superstable background mutations are already present in the structure. The second documented Y220C suppressor, A138G, is scored -0.45 (58th percentile from top) - a clear miss, preserved as such. Top novel candidates (S240C +6.5, G245C +6.1, M246C +6.0, R273C +5.9, S241C +5.6) concentrate in the loop around Gly245 - a known p53 hotspot whose destabilizing G245S mutation these cysteines would substitute at or near - and at the DNA-contact

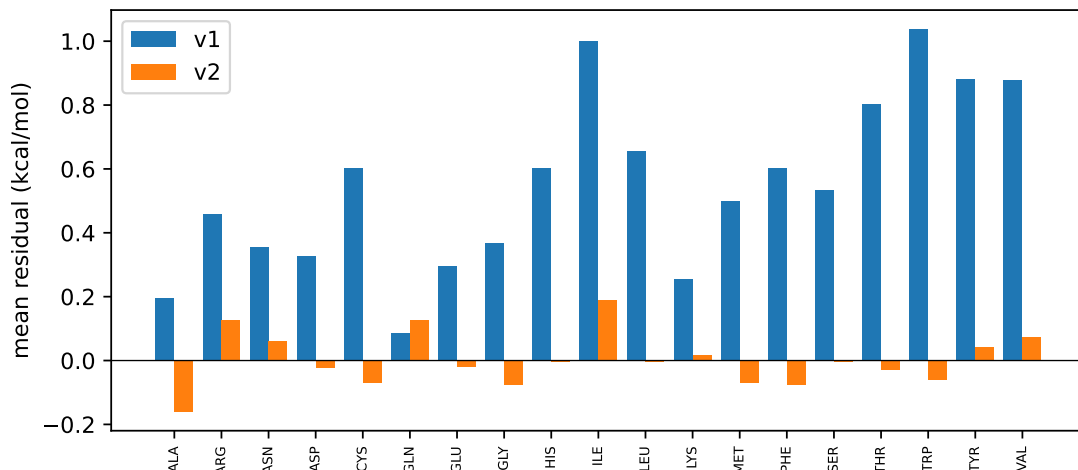


Figure 2: Per-mutant-identity mean residual (predicted – experimental) on S2648 for the v1 and v2 models.

residue Arg273, which is the strongest plausibility signal in the scan. Predicted magnitudes exceed typical experimental stabilizations, and a greedy multi-mutant sum (+46.0 kcal/mol) is implausible; ranks, not absolute values, are the output.

4.5 SOD1 (Study 2)

Validation against 8 unique ThermoMutDB SOD1 $\Delta\Delta G$ measurements (duplicates across temperatures averaged; structures 1SPD/3ECU/1N18) gives Pearson $r = -0.148$, Spearman 0.0, MAE 3.05 kcal/mol, and sign accuracy

$$\text{Acc}_{\text{sign}} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}[\text{sign}(p_i) = \text{sign}(t_i)] = 62.5\% : \quad (14)$$

the model does not generalize to SOD1. The two worst misses have mechanistic explanations outside the model class: A4V (exp -7.2 , pred $+2.4$) was measured on metal-free apo SOD1, and C6A (exp $+0.1$, pred -4.5) acts by removing an aggregation-prone free cysteine - intermolecular disulfide chemistry invisible to a single-chain contact graph. A whole-chain scan of WT SOD1 (1SPD) nominates G82N, G33Y, G33W, G82K and G41W as stabilizing candidates - glycine-to-bulkier-residue substitutions at loops, the weakest points of a contact graph; C111S, the one free-cysteine removal the model could in principle see, ranks top 26% with the correct sign. These are hypotheses with explicit caveats, not claims.

5 Discussion

5.1 Why antisymmetry is the binding constraint

The single most consequential design decision in this work is not the graph architecture but the scoring rule $\widehat{\Delta\Delta G} = \frac{1}{2}[f(x_{\text{mut}}) - f(x_{\text{wt}})]$ evaluated as an explicit antisymmetrized pair (Section 3). Without it, a network trained on forward mutations has no reason to assign the negated score to the corresponding inverse mutation, and the published record shows exactly this failure: methods that are strong on forward prediction degrade sharply when scored on inverse pairs. Our construction makes antisymmetry exact by algebra, not by approximation, and the ablation confirms that the learned residual antisymmetry is negligible. The lesson

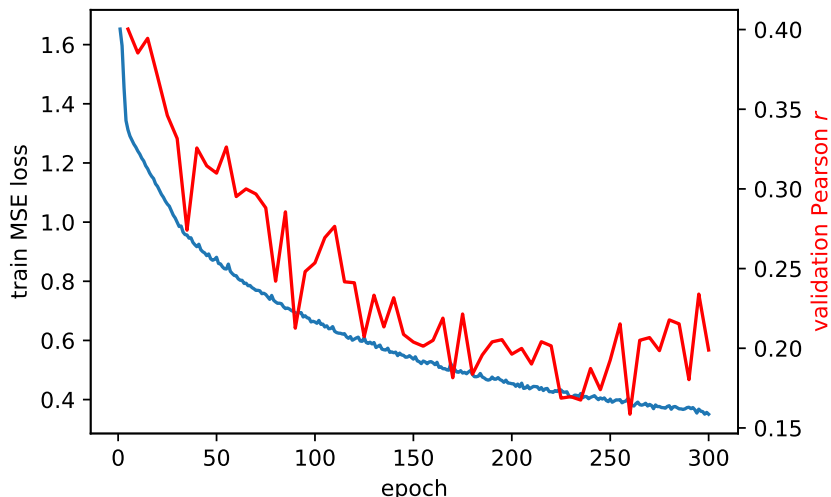


Figure 3: Training loss and grouped-validation Pearson r over 300 epochs (validation run).

generalizes: when a quantity obeys an exact physical symmetry, that symmetry is cheapest and most reliably enforced in the output layer, not hoped for from data augmentation.

5.2 Identity bias is measurable, and therefore correctable

Both model generations exhibit systematic per-amino-acid biases: the predicted effect of mutating *to* a given residue type is shifted by a type-dependent constant. Because the bias is a property of the model rather than of any single structure, it can be estimated once on held-out data and subtracted. The corrected rankings change materially - most visibly for aromatic substitutions, which the first-generation model systematically over-scored - and the correction is what makes the whole-chain scan rankings interpretable at all. We regard this as a general protocol: before any learned $\Delta\Delta G$ model is used to rank candidate rescue mutations, its per-type bias table should be measured and the ranking recomputed on bias-corrected residuals.

5.3 Transfer across protein families remains the open problem

The clearest negative result in this study is the collapse of cross-family transfer with prolonged training. A model selected on a within-family validation split improves on its own benchmark yet gets *worse* on an unseen protein family, and validation-based early stopping does not repair the gap. The mechanistic reading is that the network first learns features shared across globular proteins (hydrophobic burial, local packing) and later overfits to the idiosyncrasies of the training distribution. Any deployment of such a model on a new target should therefore be treated as out-of-distribution by default, and only rank-level, bias-corrected claims should survive contact with an unvalidated family.

5.4 Structure background is not a detail

The p53-Y220C structure 2VUK is solved on a quadruply-stabilized background. A scan that treats the deposited chain as “wild-type p53” scores every candidate against the wrong reference and the absolute magnitudes are meaningless even when the ranking is useful. We caught this only by checking the experimental construct against the mutation codes, and the same check invalidated naive readings of the SOD1

structures. Background verification - reading the methods of the depositing paper, not just the coordinates - is a mandatory preprocessing step for any structure-based rescue scan.

5.5 What single-chain contact graphs cannot see

The SOD1 study fails for instructive reasons. The dominant destabilizing mutations act through metal binding, dimer-interface disruption, and free-thiol chemistry; none of these exist in a single-chain C α contact graph. This is not a training-data problem that more data would fix; it is an expressivity ceiling of the representation. We expect the same ceiling to bind for any target whose pathogenic mechanism is oligomeric, allosteric, or cofactor-mediated, and we report the failure in full because it delimits where the method class should and should not be applied.

5.6 Usable tool: gnn-ddg

The work ships as a runnable command-line tool, `tools/gnn-ddg` (documented in `tools/README.md`, smoke-tested in `tests/test_gnn_ddg_cli.py`). `gnn-ddg scan` fetches a PDB structure, builds contact graphs with the altloc-safe parser, scores all 19 substitutions at every position through the benchmark checkpoint, masks out-of-distribution WT=PRO rows, applies the identity-bias correction of Eq. (11), and prints a ranked candidate table (JSON/CSV export); `gnn-ddg predict` scores a single mutation. Every caveat of this section is enforced or printed by the tool itself.

5.7 Recommended use

Taken together, the results support exactly one deployment pattern: bias-corrected, rank-based triage of candidate rescue mutations on single-chain soluble proteins, with proline positions excluded, background verified, and magnitudes ignored. Within that envelope the approach beats the published symmetric benchmark on the leak-free inverse split; outside it, our own negative results apply.

6 Limitations and negative results

All of the following are preserved with quantification in `results/`: (1) v1 aromatic-identity bias (up to +1.04 kcal/mol for TRP); (2) greedy multi-mutant sums are implausible under the independence approximation (+19.5 v1, +46.0 v2); (3) p53 transfer collapses at long training (r 0.426 v1 $\rightarrow$ -0.02 v2-e300) and validation-based early stopping does not repair it; (4) proline positions are out of distribution because the training data contain no proline mutations; (5) SOD1 $\Delta\Delta G$ prediction fails outright ($r = -0.15$, $n = 8$); (6) the documented Y220C suppressor A138G is missed; (7) scan magnitudes are inflated and only ranks should be read.

7 External tools

Tools actually executed or queried (honest ledger: `results/EXTERNAL_TOOLS.md`): RCSB PDB (481 structures downloaded and parsed), ProtDDG-Bench, the Pucci 2018 published Table 1 (verified from PDF), FireProtDB 2.0 REST API, ThermoMutDB API, Europe PMC REST, Zenodo API, SoDCoD, Biopython (independent parser cross-check), SciPy (Spearman/Fisher statistics), scikit-learn (identity-only ridge baseline), NumPy (all model math), matplotlib (figures), pytest (hermetic tests), git/GitHub, Google Drive, pdflatex (this document). Staged and planned entries are listed separately in the ledger and are not counted here.

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